

## 1. General Description

The EM74LVC86A is a quad 2-input EXCLUSIVE-OR gate. Inputs can be driven from either 3.3 V or 5 V devices. This feature allows the use of these devices as translators in mixed 3.3 V and 5 V applications.

Schmitt-trigger action at all inputs makes the circuit tolerant of slower input rise and fall times.

This device is fully specified for partial power down applications using  $I_{OFF}$ . The  $I_{OFF}$  circuitry disables the output, preventing the potentially damaging backflow current through the device when it is powered down.

## 2. Features and Benefits

- Wide supply voltage range from 1.2 V to 5.5 V
- Overvoltage tolerant inputs to 5.5 V
- CMOS low power dissipation
- Latch-up performance exceeds 100 mA
- Direct interface with TTL levels
- $I_{OFF}$  circuitry provides partial Power-down mode operation
- Complies with JEDEC standard:
  - JESD8-7 (1.65 V to 1.95 V)
  - JESD8-5 (2.3 V to 2.7 V)
  - JESD8C (2.7 V to 3.6 V)
  - JESD36 (4.5 V to 5.5 V)
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 6000 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

## EM74LVC86A

Quad 2-input EXCLUSIVE gate

### 3. Ordering Information

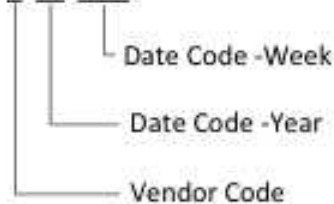
Table 1. Ordering information

| Type number  | Topside marking | Package   |                                                                           |          |
|--------------|-----------------|-----------|---------------------------------------------------------------------------|----------|
|              |                 | Name      | Description                                                               | Quantity |
| EM74LVC86AD  | LVC86A<br>XYWW  | SOP-14L   | plastic small outline package; 14 leads;<br>body width 3.9 mm             | 3000     |
| EM74LVC86APW | LVC86A<br>XYWW  | TSSOP-14L | plastic thin shrink small outline package; 14<br>leads; body width 4.4 mm | 3000     |

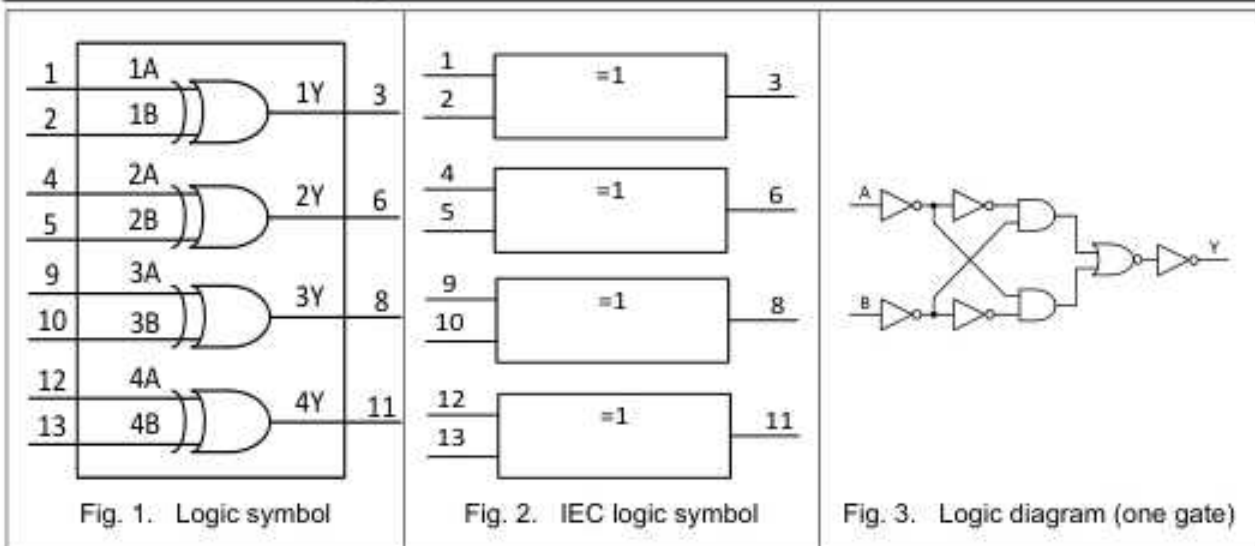
#### MARKING INFORMATION

NOTE: XYWW = Vendor Code and Date Code.

X YY WW



### 4. Function Diagram



## 5. Pinning Information

### 5.1. Pinning

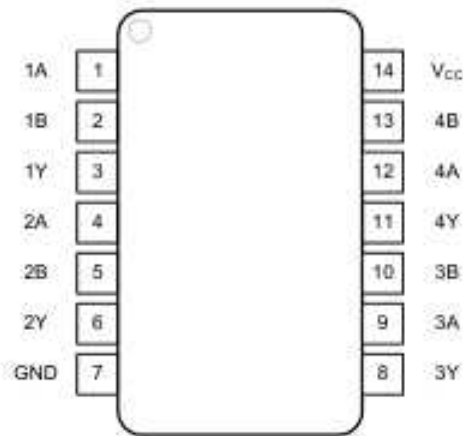


Fig. 4. Top view pin configuration SOP and TSSOP

### 5.2. Pin description

Table 2. Pin description

| Symbol          | Pin          | Description    |
|-----------------|--------------|----------------|
| 1A, 2A, 3A, 4A  | 1, 4, 9, 12  | Data input     |
| 1B, 2B, 3B, 4B  | 2, 5, 10, 13 | Data input     |
| 1Y, 2Y, 3Y, 4Y  | 3, 6, 8, 11  | Data output    |
| GND             | 7            | Ground (0V)    |
| V <sub>CC</sub> | 14           | Supply voltage |

## 6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

| Input nA | Input nB | Output nY |
|----------|----------|-----------|
| L        | L        | L         |
| L        | H        | H         |
| H        | L        | H         |
| H        | H        | L         |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                          | Min  | Max            | Unit |
|-----------|-------------------------|-------------------------------------|------|----------------|------|
| $V_{CC}$  | supply voltage          |                                     | -0.5 | 6.5            | V    |
| $I_{IK}$  | input clamping current  | $V_I < 0$ V                         | -50  |                | mA   |
| $V_I$     | input voltage           | [1]                                 | -0.5 | 6.5            | V    |
| $I_{OK}$  | output clamping current | $V_O > V_{CC}$ or $V_O < 0$ V       |      | $\pm 50$       | mA   |
| $V_O$     | output voltage          | Active mode [1]                     | -0.5 | $V_{CC} + 0.5$ | V    |
|           |                         | Power-down mode; $V_{CC} = 0$ V [1] | -0.5 | 6.5            | V    |
| $I_O$     | output current          | $V_O = 0$ V to $V_{CC}$             |      | $\pm 50$       | mA   |
| $I_{CC}$  | supply current          |                                     |      | 100            | mA   |
| $I_{GND}$ | ground current          |                                     | -100 |                | mA   |
| $P_{TOT}$ | total power dissipation | $T_{amb} = -40$ °C to $+125$ °C     |      | 500            | mW   |
| $T_{stg}$ | storage temperature     |                                     | -65  | 150            | °C   |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions                      | Min  | Max      | Unit |
|---------------------|-------------------------------------|---------------------------------|------|----------|------|
| $V_{CC}$            | supply voltage                      |                                 | 1.65 | 5.5      | V    |
|                     |                                     | functional                      | 1.2  |          | V    |
| $V_I$               | input voltage                       |                                 | 0    | 5.5      | V    |
| $V_O$               | output voltage                      | Active mode                     | 0    | $V_{CC}$ | V    |
|                     |                                     | Power-down mode; $V_{CC} = 0$ V | 0    | 5.5      | V    |
| $T_{amb}$           | ambient temperature                 |                                 | -40  | 125      | °C   |
| $\Delta t/\Delta V$ | input transition rise and fall rate | $V_{CC} = 1.65$ V to $2.7$ V    | 0    | 20       | ns/V |
|                     |                                     | $V_{CC} = 2.7$ V to $3.6$ V     | 0    | 10       | ns/V |
|                     |                                     | $V_{CC} = 4.5$ V to $5.5$ V     | 0    | 5        | ns/V |

## 9.Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

| Symbol   | Parameter                 | Conditions                                                                   | -40 °C to +85 °C |           |              | -40 °C to +125 °C |              | Unit          |
|----------|---------------------------|------------------------------------------------------------------------------|------------------|-----------|--------------|-------------------|--------------|---------------|
|          |                           |                                                                              | Min              | Typ[1]    | Max          | Min               | Max          |               |
| $V_{IH}$ | HIGH-level input voltage  | $V_{CC} = 1.2\text{ V}$                                                      | 1.08             |           |              | 1.08              |              | V             |
|          |                           | $V_{CC} = 1.65\text{ V to }1.95\text{ V}$                                    | $0.65V_{CC}$     |           |              | $0.65V_{CC}$      |              | V             |
|          |                           | $V_{CC} = 2.3\text{ V to }2.7\text{ V}$                                      | 1.7              |           |              | 1.7               |              | V             |
|          |                           | $V_{CC} = 2.7\text{ V to }3.6\text{ V}$                                      | 2.0              |           |              | 2.0               |              | V             |
|          |                           | $V_{CC} = 4.5\text{ V to }5.5\text{ V}$                                      | $0.7V_{CC}$      |           |              | $0.7V_{CC}$       |              | V             |
| $V_{IL}$ | LOW-level input voltage   | $V_{CC} = 1.2\text{ V}$                                                      |                  |           | 0.12         |                   | 0.12         | V             |
|          |                           | $V_{CC} = 1.65\text{ V to }1.95\text{ V}$                                    |                  |           | $0.35V_{CC}$ |                   | $0.35V_{CC}$ | V             |
|          |                           | $V_{CC} = 2.3\text{ V to }2.7\text{ V}$                                      |                  |           | 0.7          |                   | 0.7          | V             |
|          |                           | $V_{CC} = 2.7\text{ V to }3.6\text{ V}$                                      |                  |           | 0.8          |                   | 0.8          | V             |
|          |                           | $V_{CC} = 4.5\text{ V to }5.5\text{ V}$                                      |                  |           | $0.3V_{CC}$  |                   | $0.3V_{CC}$  | V             |
| $V_{OH}$ | HIGH-level output voltage | $V_I = V_{IH}\text{ or }V_{IL}$                                              |                  |           |              |                   |              |               |
|          |                           | $I_O = -100\text{ }\mu\text{A};$<br>$V_{CC} = 1.65\text{ V to }5.5\text{ V}$ | $V_{CC} - 0.1$   |           |              | $V_{CC} - 0.1$    |              | V             |
|          |                           | $I_O = -4\text{ mA}; V_{CC} = 1.65\text{ V}$                                 | 1.2              |           |              | 1.05              |              | V             |
|          |                           | $I_O = -8\text{ mA}; V_{CC} = 2.3\text{ V}$                                  | 1.9              |           |              | 1.7               |              | V             |
|          |                           | $I_O = -12\text{ mA}; V_{CC} = 2.7\text{ V}$                                 | 2.2              |           |              | 2.05              |              | V             |
|          |                           | $I_O = -24\text{ mA}; V_{CC} = 3.0\text{ V}$                                 | 2.4              |           |              | 2.25              |              | V             |
|          |                           | $I_O = -32\text{ mA}; V_{CC} = 4.5\text{ V}$                                 | 3.8              |           |              | 3.5               |              | V             |
| $V_{OL}$ | LOW-level output voltage  | $V_I = V_{IH}\text{ or }V_{IL}$                                              |                  |           |              |                   |              |               |
|          |                           | $I_O = 100\text{ }\mu\text{A};$<br>$V_{CC} = 1.65\text{ V to }5.5\text{ V}$  |                  |           | 0.10         |                   | 0.10         | V             |
|          |                           | $I_O = 4\text{ mA}; V_{CC} = 1.65\text{ V}$                                  |                  |           | 0.45         |                   | 0.65         | V             |
|          |                           | $I_O = 8\text{ mA}; V_{CC} = 2.3\text{ V}$                                   |                  |           | 0.30         |                   | 0.45         | V             |
|          |                           | $I_O = 12\text{ mA}; V_{CC} = 2.7\text{ V}$                                  |                  |           | 0.40         |                   | 0.60         | V             |
|          |                           | $I_O = 24\text{ mA}; V_{CC} = 3.0\text{ V}$                                  |                  |           | 0.55         |                   | 0.80         | V             |
|          |                           | $I_O = 32\text{ mA}; V_{CC} = 4.5\text{ V}$                                  |                  |           | 0.55         |                   | 0.80         | V             |
| $I_i$    | input leakage current     | $V_I = 5.5\text{ V or GND};$<br>$V_{CC} = 0\text{ V to }5.5\text{ V}$        |                  | $\pm 0.1$ | $\pm 5$      |                   | $\pm 20$     | $\mu\text{A}$ |

# EM74LVC86A

Quad 2-input EXCLUSIVE gate

| Symbol          | Parameter                 | Conditions                                                                                                         | -40 °C to +85 °C |           |          | -40 °C to +125 °C |          | Unit          |
|-----------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|------------------|-----------|----------|-------------------|----------|---------------|
|                 |                           |                                                                                                                    | Min              | Typ[1]    | Max      | Min               | Max      |               |
| $I_{off}$       | power-off leakage current | $V_{CC} = 0\text{ V}$ ;<br>$V_I$ or $V_O = 5.5\text{ V}$                                                           |                  | $\pm 0.1$ | $\pm 10$ |                   | $\pm 20$ | $\mu\text{A}$ |
| $I_{CC}$        | supply current            | $V_{CC} = 1.65\text{ V to }5.5\text{ V}$ ;<br>$V_I = 5.5\text{ V or GND}$ ; $I_O = 0\text{ A}$                     |                  | 0.01      | 10       |                   | 40       | $\mu\text{A}$ |
| $\Delta I_{CC}$ | additional supply current | per input pin ;<br>$V_{CC} = 2.3\text{ V to }5.5\text{ V}$ ;<br>$V_I = V_{CC} - 0.6\text{ V}$ ; $I_O = 0\text{ A}$ |                  | 0.2       | 500      |                   | 5000     | $\mu\text{A}$ |
| $C_i$           | input capacitance         | $V_{CC} = 3.3\text{ V}$ ;<br>$V_I = \text{GND to }V_{CC}$                                                          |                  | 4         |          |                   |          | pF            |

[1] All typical values are measured at  $V_{CC} = 3.3\text{ V}$  and  $T_{amb} = 25\text{ °C}$ .

## 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6.

| Symbol      | Parameter                     | Conditions                                | -40 °C to +85 °C |        |      | -40 °C to +125 °C |      | Unit |
|-------------|-------------------------------|-------------------------------------------|------------------|--------|------|-------------------|------|------|
|             |                               |                                           | Min              | Typ[1] | Max  | Min               | Max  |      |
| $t_{pd}$    | propagation delay             | nA, nB to nY; see Fig. 5 [2]              |                  |        |      |                   |      |      |
|             |                               | $V_{CC} = 1.2V$                           |                  | 36     |      |                   |      | ns   |
|             |                               | $V_{CC} = 1.65 V$ to $1.95 V$             | 3.4              | 11.4   | 22   | 3.4               | 22.5 | ns   |
|             |                               | $V_{CC} = 2.3 V$ to $2.7 V$               | 2.7              | 6.3    | 10.5 | 2.7               | 11   | ns   |
|             |                               | $V_{CC} = 3.0 V$ to $3.6 V$               | 2.0              | 4.6    | 7.5  | 2.0               | 8.0  | ns   |
|             |                               | $V_{CC} = 4.5 V$ to $5.5 V$               | 1.5              | 3.3    | 5    | 1.5               | 5.5  | ns   |
| $t_{sk(o)}$ | output skew time              | $V_{CC} = 3.0 V$ to $3.6 V$ [3]           |                  |        | 1.0  |                   | 1.5  | ns   |
| $C_{PD}$    | power dissipation capacitance | per gate ;<br>$V_I = GND$ to $V_{CC}$ [4] |                  |        |      |                   |      |      |
|             |                               | $V_{CC} = 1.65 V$ to $1.95 V$             |                  | 12.7   |      |                   |      | pF   |
|             |                               | $V_{CC} = 2.3 V$ to $2.7 V$               |                  | 13.4   |      |                   |      | pF   |
|             |                               | $V_{CC} = 3.0 V$ to $3.6 V$               |                  | 14.3   |      |                   |      | pF   |
|             |                               | $V_{CC} = 4.5 V$ to $5.5 V$               |                  | 16.2   |      |                   |      | pF   |

[1] Typical values are measured at  $T_{AMB} = 25\text{ °C}$  and  $V_{CC} = 1.2V, 1.8 V, 2.5 V, 3.3 V$  and  $5 V$  respectively.

[2]  $t_{pd}$  is the same as  $t_{PLH}$  and  $t_{PLL}$ .

[3] Skew between any two outputs of the same package switching in the same direction. This parameter is guaranteed by design.

[4]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu W$ ).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$  where:

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

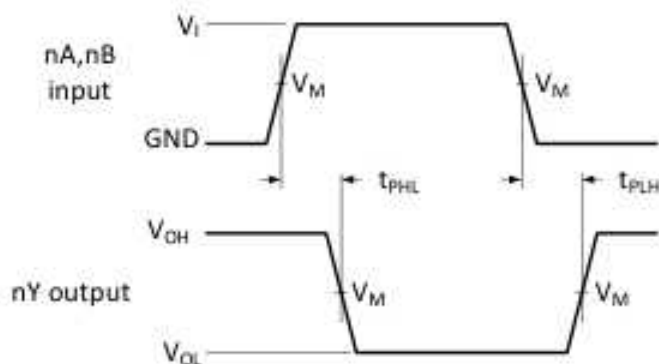
$C_L$  = output load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

$\sum (C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

## 10.1. Waveforms and test circuit



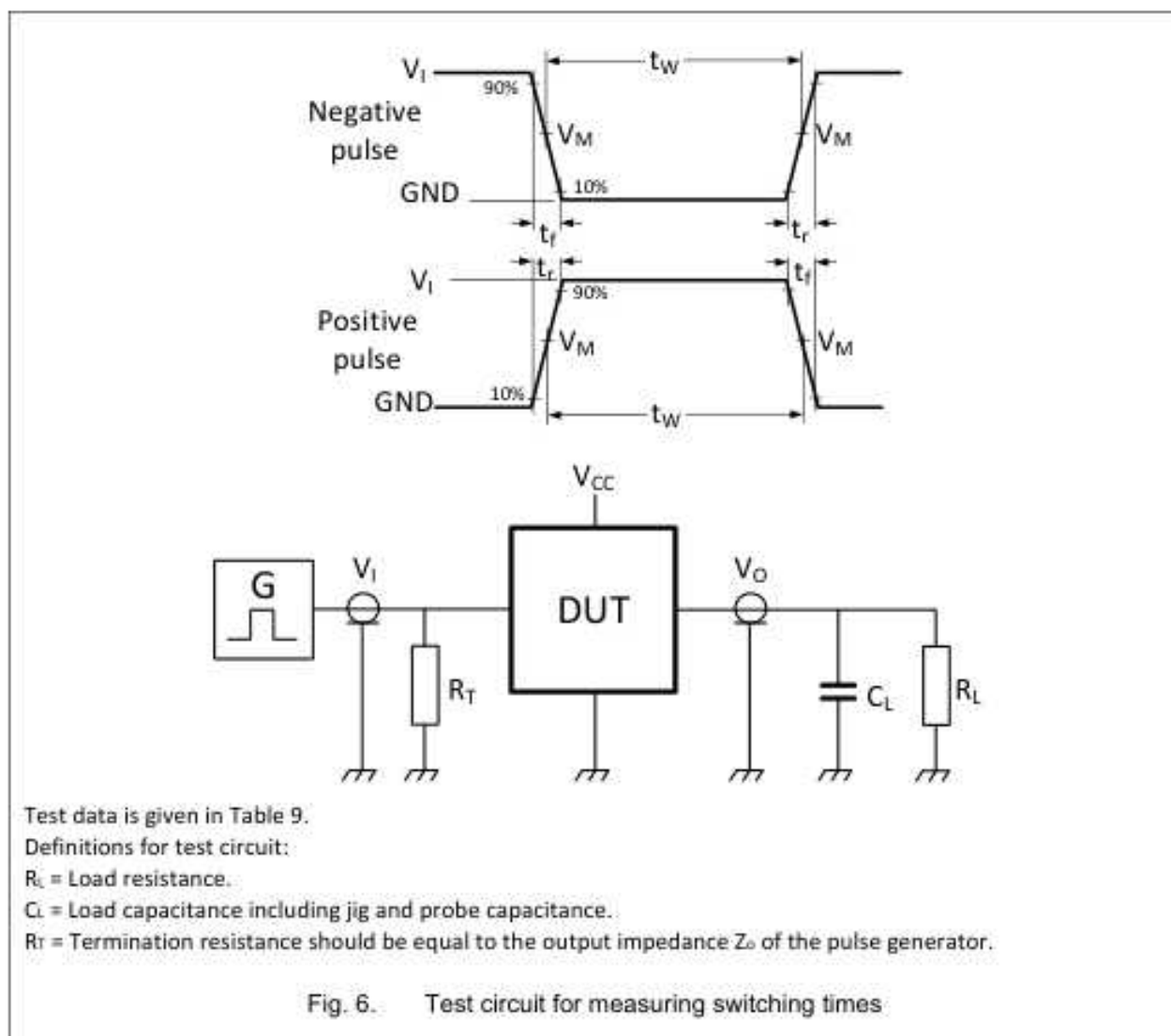
Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 5. The input nA, nB to output nY propagation delays

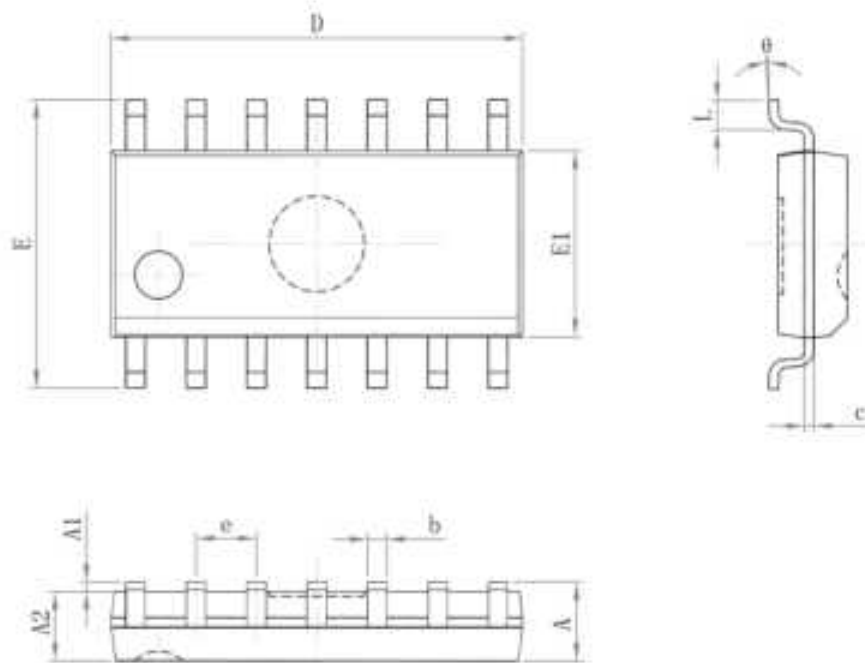
**Table 8. Measurement points**

| Supply voltage   | Input       | Output      |
|------------------|-------------|-------------|
| $V_{CC}$         | $V_M$       | $V_M$       |
| 1.2 V            | $0.5V_{CC}$ | $0.5V_{CC}$ |
| 1.65 V to 1.95 V | $0.5V_{CC}$ | $0.5V_{CC}$ |
| 2.3 V to 2.7 V   | $0.5V_{CC}$ | $0.5V_{CC}$ |
| 3.0 V to 3.6 V   | 1.5 V       | 1.5 V       |
| 4.5 V to 5.5 V   | $0.5V_{CC}$ | $0.5V_{CC}$ |


**Table 9. Test data**

| Supply voltage   | Input    |               | Load  |              |
|------------------|----------|---------------|-------|--------------|
| $V_{CC}$         | $V_I$    | $t_r = t_f$   | $C_L$ | $R_L$        |
| 1.2 V            | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ |
| 1.65 V to 1.95 V | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ |
| 2.3 V to 2.7 V   | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ |
| 3.0 V to 3.6 V   | 3 V      | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ |
| 4.5 V to 5.5 V   | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ |

## 11. Package Outline

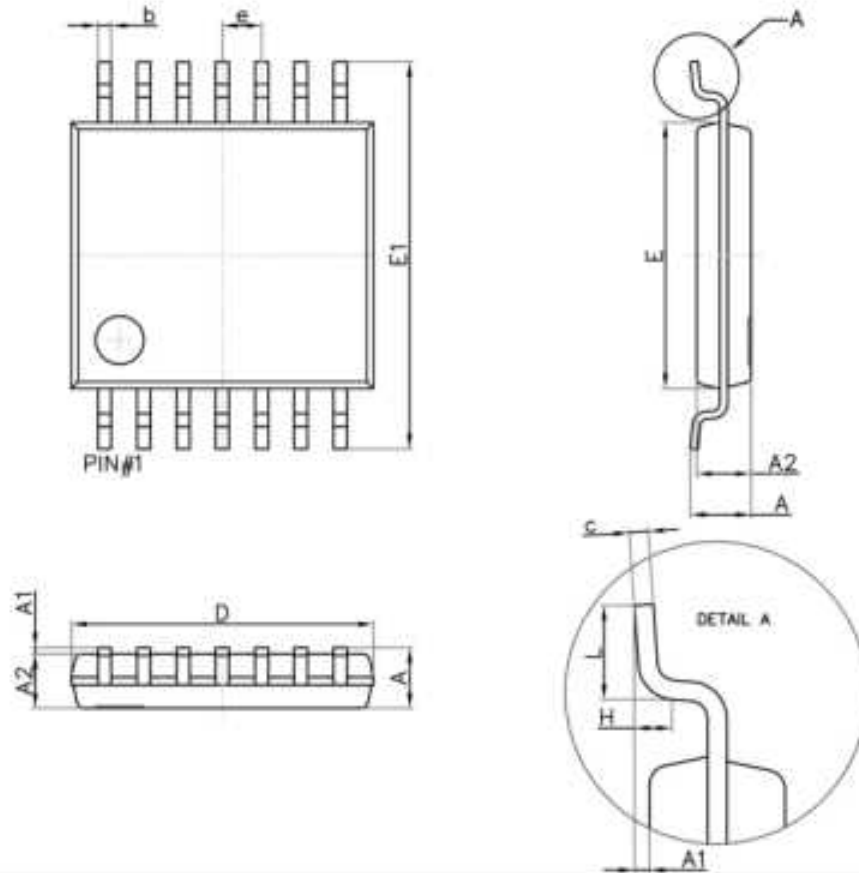
**SOP-14L**


| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min                       | Max   | Min                  | Max   |
| A      | —                         | 1.750 | —                    | 0.069 |
| A1     | 0.100                     | 0.250 | 0.004                | 0.010 |
| A2     | 1.250                     | —     | 0.049                | —     |
| b      | 0.310                     | 0.510 | 0.012                | 0.020 |
| c      | 0.100                     | 0.250 | 0.004                | 0.010 |
| D      | 8.450                     | 8.850 | 0.333                | 0.348 |
| E      | 5.800                     | 6.200 | 0.228                | 0.244 |
| E1     | 3.800                     | 4.000 | 0.150                | 0.157 |
| e      | 1.270(BSC)                |       | 0.050(BSC)           |       |
| L      | 0.400                     | 1.270 | 0.016                | 0.050 |
| theta  | 0°                        | 8°    | 0°                   | 8°    |

# EM74LVC86A

Quad 2-input EXCLUSIVE gate

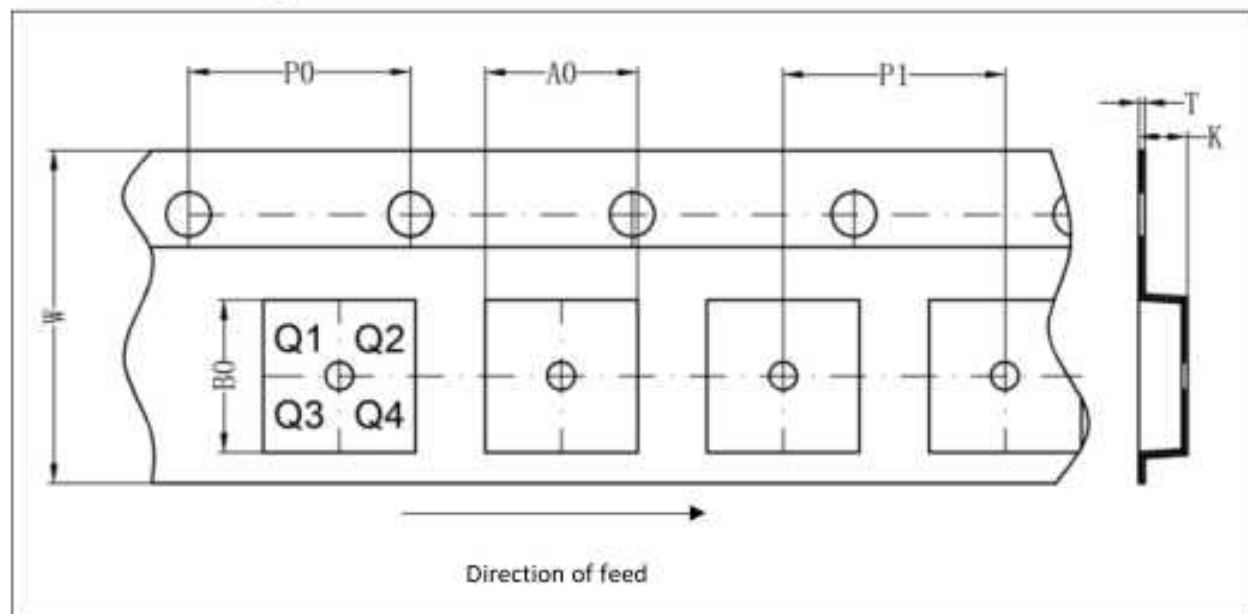
## TSSOP-14L



| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| D      | 4.900                     | 5.100 | 0.193                | 0.201 |
| E      | 4.300                     | 4.500 | 0.169                | 0.177 |
| b      | 0.190                     | 0.300 | 0.007                | 0.012 |
| c      | 0.090                     | 0.200 | 0.004                | 0.008 |
| E1     | 6.250                     | 6.550 | 0.246                | 0.258 |
| A      | —                         | 1.200 | —                    | 0.047 |
| A2     | 0.800                     | 1.000 | 0.031                | 0.039 |
| A1     | 0.050                     | 0.150 | 0.002                | 0.006 |
| e      | 0.65 (BSC)                |       | 0.026 (BSC)          |       |
| L      | 0.500                     | 0.700 | 0.020                | 0.028 |
| H      | 0.25(TYP)                 |       | 0.01(TYP)            |       |
| θ      | 1°                        | 7°    | 1°                   | 7°    |

## 12. Tape and Reel Information

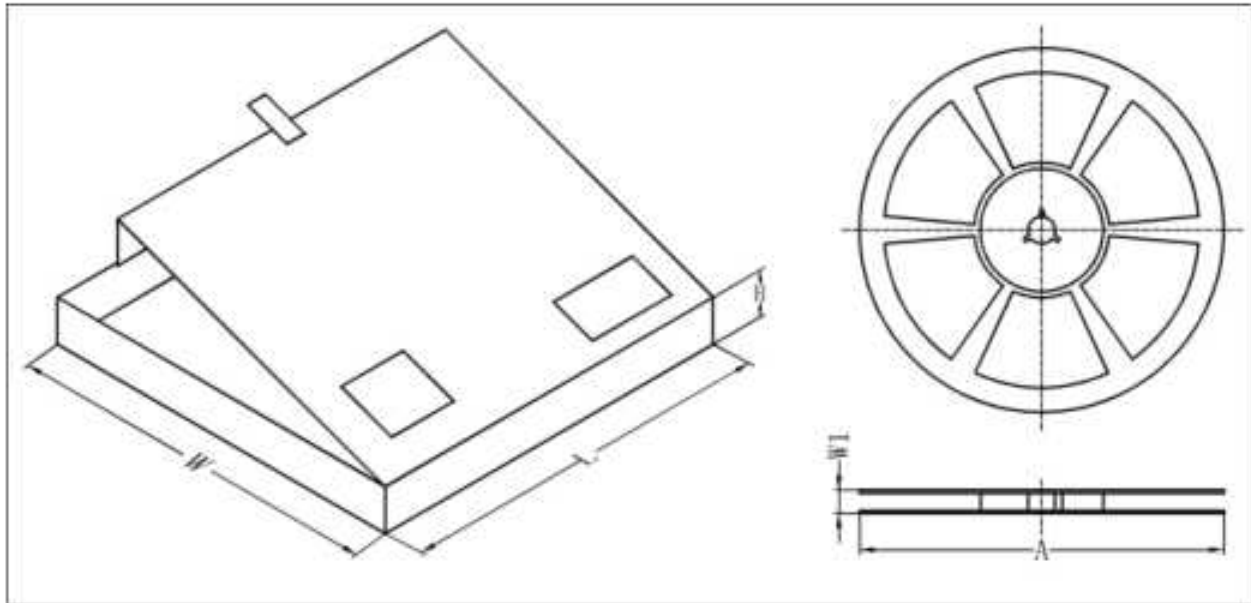
### 12.1. Carrier tape dimensions



**Table 10. Carrier tape dimensions**

| Package version | A0(mm) | B0(mm) | K0(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|--------|-------|--------|-------|--------|-------|
| SOP-14L         | 6.35   | 9.1    | 1.9    | 0.22  | 8      | 16    | 4      | Q1    |
| TSSOP-14L       | 6.7    | 5.45   | 1.6    | 0.25  | 8      | 12    | 4      | Q1    |

## 12.2. Reel and box dimensions



**Table 11. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | MPQ (pcs) | Reels per box | Outer box dimensions L×W×H(mm) |
|-----------------|-----------------|-----------------------|--------------------|-----------|---------------|--------------------------------|
| SOP-14L         | D               | 330                   | 22.4               | 3000      | 1             | 358x340x50                     |
| TSSOP-14L       | PW              | 330                   | 18.4               | 3000      | 1             | 358x340x50                     |

## 13. Abbreviations

Table 12. Abbreviations

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |
| TTL     | Transistor-Transistor Logic             |

## 14. Revision History

Table 13. Revision history

| Document ID         | Release Date                                                                                                                                                                   | Data sheet status | Change notice | Supersedes          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|---------------------|
| EM74LVC86A Rev. 1.1 | Dec 20, 2024                                                                                                                                                                   | Product datasheet |               | EM74LVC86A Rev. 1.0 |
| Modifications:      | <ul style="list-style-type: none"><li>• Table 1: Added topside marking information.</li><li>• Table 5 updated.</li><li>• Section 12 added tape and reel information.</li></ul> |                   |               |                     |
| EM74LVC86A Rev. 1.0 | Aug 30, 2024                                                                                                                                                                   | Product datasheet |               |                     |